Veritas Vision

Multimodal Phishing Detector

Reads the page the way a human victim sees it - and catches phishing URL scanners cannot see.

Google Gemini Hackathon | RUVERSE TechFest | 23 Oct 2026

Team: Grieving Souls | Presenter: Anshika Tiwari | Prize Pool: INR 40,000

Repo: github.com/Argyrys/grievingsouls-GGH

The Problem

Phishing still begins with one click

- Phishing is the #1 initial access vector for breaches and costs consumers and enterprises billions of dollars every year.
- Attackers clone login pages pixel-for-pixel and hide the forgery where URL scanners cannot see: inside <canvas> elements, SVG images, and domains one character off the real brand.
- Modern defenses scan URLs. Attackers now hide phishing in the pixels - so URL-only tools miss it entirely.
- Who it affects: every browser user, every bank customer, every employee signing into a work portal.

Our Solution

Veritas Vision sees phishing like a human does

- Three-layer detector: lexical fast-tier -> multimodal vision -> structured verdict.
- Layer 1 - Lexical fast-tier: pure-function URL scoring blocks obvious forgeries in milliseconds - zero screenshots, zero Gemini calls.
- Layer 2 - Multimodal vision: headless Chromium renders the page exactly as a victim sees it; Gemini reads brand trade dress and text baked into canvas/SVG. The decision is the mismatch between brand UI and host domain.
- Layer 3 - Verdict: structured JSON (is_phishing, confidence_score, brand, signals, block/warn/allow) shown on a dashboard and a Chrome extension badge.

How It Works

Four steps, end to end

1. Target	A URL is submitted via the dashboard or the Chrome extension popup.
2. Fast tier	The lexical engine scores the URL first; conclusive forgeries block instantly without rendering anything.
3. Vision check	Playwright captures a 1280x800 screenshot; Gemini analyzes the image + URL and returns a machine-validated JSON verdict.
4. Surface	Dashboard shows verdict, confidence gauge, extracted text, and the screenshot. Extension sets toolbar badge PH / ?? / OK.

Lexical Fast-Tier (built today)

Zero network. Zero quota. Runs before rendering.

- Signals: leetspeak (paypa1), brand token on an untrusted domain (microsoft-login.com), punycode homoglyphs, raw IP hosts, '@' credential tricks, suspicious TLDs (.xyz, .top, .icu), credential keywords on unrelated hosts.
- Threshold: risk >= 0.75 = instant block; registered brand domain = instant allow; anything ambiguous - including localhost - escalates to multimodal vision.
- Live result: https://paypa1.com blocks in 23 ms, spending 0 Gemini calls.
- Why it matters: survives the free Gemini quota (20 calls/day per model) by spending calls only where vision is actually needed.

Live Demo

Three scans, ninety seconds

https://paypa1.com	**BLOCK**	**conf 0.92**	**lexical**	0 ms, 0 Gemini calls
canvas-cloaked Microsoft page	**BLOCK**	**conf 0.99**	**multimodal**	Gemini reads brand text hidden in <canvas>
legit portal mock	**ALLOW**	**conf 0.1**	**multimodal**	Judges the mismatch, not the logo

Chrome extension: on any tab, one click sets the toolbar badge PH (phishing), ?? (suspicious) or OK (safe).

Tech Stack

Picked for speed and free-tier accessibility

Backend	Browser	AI	Frontend
FastAPI · Python · Pydantic · python-dotenv	Playwright (headless Chromium, 1280x800 screenshots)	Google Gemini API · gemini-3.5-flash-lite · structured JSON output	Vanilla HT self-containe venue Wi-Fi

How Gemini Is Used

gemini-3.5-flash-lite, multimodal, structured

- Native multimodal input: destination URL + PNG screenshot, exactly as a victim sees the page.
- Every response is machine-validated: Gemini returns structured JSON via Pydantic response_schema (no fragile text parsing).
- Prompt performs a cognitive cross-reference: visual brand dress vs URL context vs the mismatch between them.
- Quota-conscious design: multi-key rotation on 429, lexical tier absorbs obvious cases for free, trusted brand domains skip the call entirely.

Impact

Who this helps, and how much

- Everyday users: protects anyone who clicks links in email or messages - no security expertise needed.
- Banks and brands: catches lookalike domains and pixel-perfect login clones impersonating them.
- Enterprises and employees: defends work-portal credentials, a top initial-access vector.
- Accessibility: runs on the free Gemini tier - usable by students, small organizations, and individuals without budget for security tools.
- Detects what URL scanners miss: canvas-cloaked text, SVG brand art, homoglyph domains.

Team - Grieving Souls

4 members · lead presenter · repo pushed before 9:00 AM checkpoint

Name	Contribution	
Rajat Sharma	**Backend & AI Lead**	FastAPI + Gemini pipeline, lexical fast-tier, capture fix
Arpita	**Security & Testing**	Threat model, mock victim pages, unit test suite
Anshika Tiwari	**Presenter & Pitch**	Pitch deck, 90-second demo narration, judge Q&A prep
Ritik Kumar	**Frontend Developer**	Dashboard UI, Chrome extension badge

Honesty under the 60% Rule

Pre-event core vs. built on the day

Pre-event MVP (<= 60%)	FastAPI backend + Playwright + Gemini multimodal pipeline, dashboard, Chrome extension, 4 mock victim pages, run.bat launcher.
Built on 23 Oct (>= 40%)	Lexical fast-tier + auto-fire pre-check, capture hardening, file-backed persistent history, offline unit test suite, 3 new mocks (PayPal, GitHub, Swiggy), presentation materials.

Commit history is intact and real: every teammate commits from their own GitHub account, no history rewriting, no .env or secrets committed.

Credits

Everything we reused, and how

- Playwright (Apache-2.0) - headless browser capture.
- Google Gemini API / Google GenAI SDK - multimodal analysis.
- FastAPI (MIT), Pydantic (MIT), python-dotenv (MIT) - backend.
- fpdf2 (LGPL-3.0) - presenter handbook generator.
- All mock pages hand-written for this project. Brands are referenced nominatively for impersonation research - no logos, fonts, or assets copied.
- Presenter handbook (11 pages): Veritas_Vision_Presenter_Handbook.pdf.
- No copied or plagiarised project, code, or design.

Close

Thank you

Problem first -> live demo -> how it works -> close.

"We stopped looking at URLs and started looking at screens."

Repo: github.com/Argyrys/grievingsouls-GGH
Live demo: run.bat -> http://127.0.0.1:8000/

Grieving Souls - Google Gemini Hackathon, 23 Oct 2026